1.

Which language is considered a predecessor to C?

A. Java
B. C++
C. BCPL
D. Python

Answer: C

2.

The language that the computer can understand is called ___.

A. High Level Language
B. Machine Language
C. System Program
D. Assembly Language

Answer : B

3.

Which of the following is NOT a feature commonly found in modern IDEs?

A. Syntax highlighting
B. Code refactoring
C. Memory chip simulation
D. Integrated terminal

Answer: C

4.

Which of the following is the correct order of the compilation steps in C?

A. Compiling, Preprocessing, Linking, Assembling
B. Preprocessing, Compiling, Assembling, Linking
C. Assembling, Preprocessing, Compiling, Linking
D. Linking, Assembling, Compiling, Preprocessing

Answer: B

5.

What is the next step to Assembling in the C program build process?

A. Preprocessing
B. Compiling
C. Linking
D. None of the above

Answer: C

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1.

What will be the output of the following C code?

```
#include <stdio.h>
int main(void)
{
    int n1 = 10;
    printf("%d", printf("%s%d", "Hello",n1));
    return 0;
}
```

A. Hello5
B. Hello107
C. 5Hello
D. Compilation error

Answer: B

2.

What will be the output of the following C code?

```
#include <stdio.h>
int main(void)
{
    printf("10\\2 = %d", 10\2);
    return 0;
}
```

A. 5
B. 10\2 = 5
C. Compilation error
D. 10\2 = 10

Answer: C

3.

What are tokens in C?

A. Only Input/output operators
B. Only Constants and strings
C. Smallest individual units in a program
D. Only keywords and identifiers

Answer: C

4.

What will be the output of the following C code?

```
#include<stdio.h>
int main(void)
{
    int a,b;
    printf("%d ",printf("%d ",scanf("%d%d",&a,&b)));
    return 0;
}
//Note  [Consider  a=20 and b=10 ]
```

A. 2 2
B. 2 1
C. 20 10
D. 1 1

Answer: A

5.

What is the purpose of %*d in scanf()?

A. Read and store an int
B. Read but ignore an int input from storage
C. Store input into pointer
D. Print int as pointer

Answer: B

1.

Assume as per C compiler sizeof(long)=8 in such case

A. sizeof(long) > sizeof(int)
B. sizeof(long) < sizeof(int)
C. sizeof(long) == sizeof(int)
D. sizeof(long) < sizeof(char)

Answer: A

2.

What is the output of this code?

```
unsigned short x = -1;
printf("%u\n", x);
```

A. -1
B. 65535
C. Compile-time error
D. Depends on compiler
E. 65536 (2^16)

Answer: B

3.

From the expression below, which is correct to get fractional division?

A. double x = 5/2;
B. double x = 5.0/2;
C. double x = (int)5/2;
D. double x = 5/ (int)2;

Answer: B

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4.

A C expression contain two operands one as signed int and unsigned int then C compiler :

A. Do not change type of any operand
B. Converts to signed int operand to unsigned int
C. It will cause to compile time error
D. It will cause to Runtime error

Answer: B

5.

Available Data types and Type modifiers in C Programming language with latest compiler size order of short, int, long long will be?

A. short < int < long long
B. long < int < short long
C. int < short < long long
D. short < long long < int

Answer: A

1.

```
#include<stdio.h>
int main(void)
{
    int a = 5, b = 2,c=2;
    a *= ++b + c++;
    printf("%d", a);
    return 0;
}
```

A. 25
B. 32
C. 30
D. Undefined behavior

Answer: A

2.

```
#include<stdio.h>
int main(void)
{
    int a = 4, b = 5;
    a <<= b = 2;
    printf("%d", a);
    return 0;
}
```

A. 16
B. 32
C. 8
D. Undefined behavior

Answer: A

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3.

```c
#include <stdio.h>
int main(void)
{
    int x = 2, y = 3;
    int z = x & y && x | y;
    printf("%d\n", z);
    return 0;
}
```

A. 0
B. 1
C. 2
D. 3

Answer: B

4.

```c
#include<stdio.h>
int main(void)
{
    int a,b,c;
    c=scanf("%d",&b) & printf("%10d",b);
    printf("%d",c);
    return 0;
}                    // Note:- consider input for a=9
```

A. 91
B. 90
C. 91
D. 9 0

Answer :B

5.

Which is not a valid assignment operator in C?

A. +=
B. *=
C. &&=
D. %=

Answer: C

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1.

```c
#include <stdio.h>
int main(void)
{
    int a = 0, b = 6;
    if (a > 0 || b / a)
        printf("Yes");
    else
        printf("No");
    return 0;
}
```

A. Yes if b divisible by a
B. No if b divisible
C. Always Yes
D. Runtime error

Answer: D

2.

```c
#include <stdio.h>
int main(void)
{
    int x = 1;
    printf("%d", x && x-- ? x : x + 2);
    return 0;
}
```

A. 1
B. 0
C. 2
D. 3

Answer: B

3.

```c
#include <stdio.h>
int main(void)
{
    int x;
    if (x = printf("Hello"))
        printf(" world");
    if (x >>= 1)
        printf("Shifted non-zero");
    else
        printf("Zero");
    printf("and now x is %d",x);
    return 0;
}
```

A. Hello worldShifted non-zero and now x is 2
B. Hello worldZero and now x is 5
C. HelloZero and now x is 2
D. worldShifted non-zero and now x is 10

Answer: A

4.

Which of the following is TRUE about nested conditional operators in C?

A. Parentheses are required for every nested ternary expression
B. Compiler throws error for nested ternary expressions
C. Nested ternary expressions work without parentheses but may reduce readability
D. Nested ternary expressions are not supported in standard C

Answer: C

5.

```c
#include <stdio.h>
int main(void)
{
    int a = 5, b = 10, c = 15;
    int result = (a > b) ? ((b > c) ? b : c) : a;
    printf("%d", result);
    return 0;
}
```

A. 5
B. 10
C. 15
D. Compile error

Answer: A

6.

What is the output of the following code?

```c
int a = 0;
int b = 2;
int c = 4;
int result = a ? b : c == 4 ? b : c;
printf("%d", result);
```

A. 0
B. 2
C. 4
D. Compile error

Answer: B

1.

What is the output of the following code?

```c
#include <stdio.h>
typedef enum
{
    X, Y, Z
} E;
typedef E EnumType;
int main(void)
{
    EnumType e = 5;
    switch(e)
    {
        case X: printf("X\n");
                break;
        case Y: printf("Y\n");
                break;
        case Z: printf("Z\n");
                break;
        default:
                printf("Default\n");
    }
    return 0;
}
```

A. X
B. Y
C. Z
D. Default

Answer: D

2.

What is the output of the following code?

```
#include <stdio.h>
enum day
{
    MON = 1, TUE, WED = 5, THU, FRI = MON + 4
};
int main(void)
{
    enum day today = THU;

    int val = today + FRI - TUE;

    printf("%d\n", val);

    return 0;
}
```

A. 9
B. 10
C. 11
D. 12

Answer: A

3.

What is the output of the following code?

```
#include <stdio.h>
int main(void)
{
    int a = 1;
    switch(a)
    {
        case 1;
                printf("One");
                break;
        default:
                printf("Default");
    }
    return 0;
}
```

A. One
B. Default
C. Compile time error
D. No output

Answer: C

4.

What is the output of the following code?

```
#include <stdio.h>
typedef enum
{
    SUNDAY, MONDAY, TUESDAY
} Day;
int main(void)
{
    Day today = MONDAY;
    switch(today)
    {
        case SUNDAY:
                printf("Weekend");
                break;
        case MONDAY:
                printf("Start of Week");
                break;
        default:    printf("Midweek");
                break;
    }
    return 0;
}
```

A. Weekend
B. Start of Week
C. Midweek
D. Compile Error

Answer: B

5.

```
#include <stdio.h>
typedef enum {RED, GREEN, BLUE} Color;
int main(void)
{
    Color c = 0 + 1 * 0 + 1 % 2;
    if (c == RED)
        printf("Red\n");
    else if (c == GREEN)
        printf("Green\n");
    else
        printf("Blue\n");
    return 0;
}
```

A. Red
B. Green
C. Blue
D. Compile time error

Answer: B

6.

What is the output of the following code?

```
#include<stdio.h>
typedef enum {MON=1, TUE, WED} Day;
int main(void)
{
    Day d = 2;
    if (d == MON)
        printf("Monday");
    else if (d == --TUE)
        printf("Tuesday");
    else
        printf("Other day");
    return 0;
}
```

A. Monday
B. Tuesday
C. Other day
D. Compile time error

Answer: D

7.

What is the output of the following code?

```
#include<stdio.h>
int main(void)
{
    int n = 5;
    switch(n % 2)
    {
        case 0:
                if (n == 0) printf("Zero");
                else printf("Even");
                    break;
        default:
                printf("Odd");
    }
}
```

A. Zero
B. Even
C. Odd
D. Compile error

Answer: C

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1.

Find the output of the following

```
#include<stdio.h>
typedef enum
 {
	EVEN, ODD
} Type;
int main(void)
{
	int i=0;
	Begin:
	i++;
	for (; i < 5; i++ )
	{
		Type t = (i % 2 == 0) ? EVEN : ODD;
		if (t == ODD)
		{
			printf("%d ", i);
			break;
			goto Begin;
		}
	}
	return 0;
}
```

A. 1

B. 0 2 4

C. 0 1 2 3 4

D. Compile time error

E. infinute loop

Answer: A

2.

Find the output of the following

```
#include<stdio.h>
typedef enum {ON=1, OFF=0} Power;
int main(void)
{
	Power state = OFF;
	int count = 0;
	do
	{
		if (state == OFF && count == 2)
			state = ON;
		printf("%d ", count);
		count++;
	}while (state == OFF);
	return 0;
}
```

A. 0 1 2

B. 0 1

C. 0 1 2 3

D. Infinite loop

E. 0 2 4

Answer: A

3.

Find the output of the following

```
#include<stdio.h>
int main(void)
{
    int i = 0;

    End:
    while (i < 3)
    {
        printf("Loop %d\t", i);
        if (i == 1)
          goto Start;
        i++;
    }
    printf(" SunBeam ");

    Start:

    printf(" End ");
    return 0;
}
```

A. Loop 0 Loop 1 SunBeam End

B. Loop 0 Loop 1 End

C. Loop 0 End

D. Compile Error

D. infinte loop

Answer: B

4.

Find the output of the following

```
#include<stdio.h>
typedef enum {START=1, STOP=5} State;
int main(void)
{
    State s;
    for (s = START; s <= STOP; s++)  {
        if (s == 3)
            continue;
        printf("%d ", s);
    }
    return 0;
}
```

A. 1 2 3 4 5

B. 1 2 4 5

C. 1 2 3 5

D. Compile time error

Answer: B

5.

Find the output of the following

```
#include<stdio.h>
int main(void)
{
    for (int i = 0; i < 5; i++) {
        if (i != 2)
            continue;
        printf("%d ", i);
    }
    return 0;
}
```

A. 0 1 2 3 4
B. 0 1 3 4
C. 0 1 2
D. 1 2 3 4
E. 2

Answer: E

1.

Choose the correct statement:

A. static variable gets memory every time the function is called
B. register variable can be accessed using pointers
C. extern variables are accessible across multiple files
D. auto variables have global scope

Answer: C

2.

Find the output of the following code:

```
#include <stdio.h>
void modify(int x)
{
    x += 10;
    printf("%d ", x);
    return;
}
int main(void)
{
    int x = 5;
    modify(x++);
    printf("%d", x);
    return 0;
}
```

A. 15 5
B. 15 15
C. 15 6
D. Compile error

Answer: C

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3.

Find the output of the following code:

```
#include <stdio.h>
void Hello();
int main(void)
{
    Helo();
    return 0;
}
void Hello()
{
    printf("Hello");
    return;
}
```

A. Compile error
B. Prints "Hello"
C. Linker error
D. Garbage output
E. Run time error

Answer: C

4.

Which of the following is true about static variables in C?

A. They are destroyed when function completes its execution
B. They are reinitialized each time
C. They will retain values across multiple calls
D. They cannot be used inside functions

Answer: C

5.

Find the output of the following code:

```
#include <stdio.h>
void test()
{
    static int a = 0;
    a++;
    printf("%d ", a);
    return;
}
int main()
{
    for (int i = 0; i < 3; i++)
    {
        test();
    }
    return 0;
}
```

A. 1 1 1
B. 1 2 3
C. 3 3 3
D. 0 1 2
E. garbage values

Answer: B

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6.

Find the output of the following code:

```
#include <stdio.h>
int abc(int i);
int main(void)
{
    int result=abc(10);
    printf("%d",--result);
    return 0;
}
int abc(int i)
{
    return i++, i-5;
}
```

A. 10
B. 9
C. 11
D. None of these

Answer: D

1.

Find the output of the code:

```
#include <stdio.h>
int multiply(int a, int b)
{
    if (b == 0)
        return 0;
    return a + multiply(a, b - 1);
}
int cube(int n)
{
    if (n == 0)
        return 0;
    return multiply(n * n, n);
}
int main(void)
{
    int n = 3;
    printf("Cube of %d is %d\n", n, cube(n));
    return 0;
}
```

A. Cube of %d is 3
B. Cube of %d is 9
C. Cube of %d is 27
D. Infinite recursion error

Answer: C

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2.

Find the output of the code:

```
#include <stdio.h>
int fun(int x)
{
    static int s = 5;
    if (x > 0)
    {   s += x;
        return fun(x - 1);
    }
    return s;
}
int main(void)
{
    printf("%d", fun(3));
    return 0;
}
```

A. 5
B. 6
C. 11
D. 14

Answer: C

3.

Find the output of the code:

```
#include <stdio.h>
void fun(int x);
int main(void)
{
    fun(3);    return 0;
}
```



```
void fun(int x)
{
    if (x > 0)
    {
        fun(x - 1);
        fun(x - 2);
        printf("%d ", x);
    }
}
```

A. 1 2 3
B. 1 2 2 3
C. 1 2 1 3
D. Too many outputs to predict

Answer: C

4.

What does this function compute?

```
int fun(int x, int y)
{
    if (y == 0)
        return 1;
    return x * fun(x, y - 1);
}
```

A. x^y
B. y^x
C. x+y
D. x-y

Answer: A

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5.

Find the output of the code:

```
#include <stdio.h>
int fun(int x)
{
    if (x <= 0)
        return 0;
    else
        return x + fun(x - 2);
}
int main(void)
{
    printf("%d", fun(5));
    return 0;
}
```

A. 9
B. 8
C. 10
D. 12

Answer: A

6.

Find the output of the code:

```
#include <stdio.h>
int fun(int x)
{
    if (x < 10)
        return x;
    return fun(x / 10) + (x % 10);
}
int main(void)
{
    printf("%d", fun(123));
    return 0;
}
```

A. 123
B. 6
C. 5
D. 3

Answer: B

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1.

Which symbol is used to declare a pointer?

A. &

B. #

C. *

D. @

Answer: C

2.

```
#include <stdio.h>
void change(int * p)
{
	*p += 5;
	p = NULL;
	return;
}
int main(void)
{
	int a = 3;
	int *ptr = &a;
	change(ptr);
	printf("%d", a);
	return 0;
}
```

A. 3

B. 8

C. 0

D. compiler error

Answer: B

3.

```
#include <stdio.h>
void square(int *const p)
{
	int temp = *p,k=temp;
	*p = temp++ * k;
	return;
}
int main(void)
{
	int x = 4;
	square(&x);
	printf("%d", x);
	return 0;
}
```

A. 20

B. 8

C. 16

D. Compile time error

Answer: C

4.

```
#include <stdio.h>
int main(void)
{
	int a = 1, b = 2;
	int *p = &a, *q = &b;
	p = q;
	*p = 10;
	printf("%d %d", a, b);
	return 0;
}
```


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A. 10 10
B. 1 10
C. 10 2
D. 1 2

Answer: B

5. If ptr is a pointer to float, what is the result of ptr + 1?

A. Address + 1 byte
B. Address + 2 bytes
C. Address + 4 bytes (on 32-bit)
D. Address + sizeof(int)

Answer: C

6.
```
#include <stdio.h>
int main(void)
{
    int a = 10;
    int *ptr = &a;
    int c = ++*ptr, ++*ptr, ++*ptr;
    ;
    printf("%d", a);
    return 0;
}
```

A. 10
B. compiler error
C. 11
D. 13

Answer: B

1. Find the output of the following:

```
char *str = "Hello";
str[0] = 'h';
printf("%s", str);
```

A. hello
B. Hello
C. Compile error
D. Segmentation fault

Answer: D

2. Find the output of the following:

```
#include<stdio.h>
void set(int *arr)    {
    for (int i = 0; i < 3; i++)
        *(arr + i) = i * 10;
}
int main(void)
{
    int a[3];
    set(a);
    printf("%d", a[2]);
    return 0;
}
```

A. 2
B. 10
C. 20
D. 30

Answer: C

3. Find the output of the following:

```
#include<stdio.h>
void mystery(int *p)
{
    *p = *p + 5;
    p = p + 2;
    *p = *p + 10;
}
int main(void)
{
    int a[] = {1, 2, 3, 4};
    mystery(a);
    printf("%d %d", a[0], a[2]);
    return 0;
}
```

A. 6 13
B. 1 12
C. 6 12
D. 1 13

Answer: A

4. Find the output of the following:

```
#include<stdio.h>
void doubleValues(int *arr, int n)
{
    for (int i = 0; i < n; i++)
        *(arr + i) = 2 * *(arr + i);
}
```



```
int main(void)
{
    int a[] = {1, 2, 3};
    doubleValues(a, 3);
    printf("%d", a[1]);
}
```

A. 2
B. 4
C. 6
D. 3

Answer: B

5. Find the output of the following:

```
#include<stdio.h>
void process(int *arr, int i, int j)
{
    int temp = ++(i+j)[arr];
}
int main(void)
{
    int a[] = {1, 2, 3};
    process(a, 0, 2);
    printf("%d, %d, %d", a[0], a[1], a[2]);
    return 0;
}
```

A. 1,2,3
B. 1,2,4
C. 2,3,4
D. Compile Time Error

Answer: B

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6.

Find the output of the following:

```
#include <stdio.h>
void f(int *p)
{
    p = p + 1;
    *p = *p + 5;
    p = p - 1;
    *p = *p * 2;
    return;
}
int main(void)
{
    int a[] = {5, 6};
    f(a);
    printf("%d %d\n", a[0], a[1]);
    return 0;
}
```

A. 5 6

B. 10 6

C. 5 11

D. 10 11

Answer: D

1.

Below function will return which value?

```
int my_strcmp(char *s1, char *s2)
{
    while (*s1 && (*s1 == *s2))
    {
        s1++; s2++;
    }
    return *s1 - *s2;
}
```

A. Always 0

B. Returns 1 or -1

C. Same as strcmp()

D. Returns length difference

Answer: C

2.

Find the output of the below code:

```
#include<stdio.h>
int main(void)
{
    char str[] = "abcd";
    printf("%c", *(str + *(str + 2) - 97));
    return 0;
}
```

A. a

B. b

C. c

D. d

Answer: C

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3.

Find the output of the below code:

```
#include <stdio.h>
#include <string.h>
int main(void)
{
	char a[10] = "sun";
	char b[10] = "sun";
	char *x =a, *y = b;
	if (strcmp(a, b))
		x++;
	else
		y++;
	y[-1]=70;
	strcat(x,strcpy(y,"east")-1);
	printf("%s",a);
}
```

A. sunFeast

B. suFeast

C. Compile Error

D. sunfeast

Answer: A

4.

Find the output of the below code:

```
#include<stdio.h>
int main(void)
{
	char str[] = "abcde";
	printf("%c", *(&str[3] + 1));
	return 0;
}
```


A. c

B. d

C. e

D. Compile error

Answer: C

5.

Find the output of the below code:

```
#include<string.h>
#include<stdio.h>
char * demo(char *s);
int main(void)
{
	char *s = "xyz";
	printf("%s", demo(s));
	return 0;
}
char * demo(char *s)
{
	return (s + strlen(s));
}
```

A, xyz

B, x

C, z

D, No Output

Answer: D

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1.

Find the output of the following code:

```
#include <stdio.h>
int main(void)
{
    char *arr[] = {"dac", "dbda"};
    printf("%c", *arr[0] + 1);
    return 0;
}
```

A. b
B. d
C. c
D. e

Answer: D

2. Find the output of the following code:

```
#include <stdio.h>
int main(void)
{
    char data[]= " DAC";  char *ptr= data;
    ptr[0] = 'K';
    printf("%s\n", ptr);
    return 0;
}
```

A. DAC
B. KDAC
C. Compilation error
D. Segmentation fault

Answer: B

3.

Find the output of the following code:

```
#include <stdio.h>
void fun(char a[][100], int rows)
{
    for (int i = 0; i < rows-1; i++)
    {
        for (int j = 0; a[i][j] != '\0'; j++)
        {
            if ((i + j) % 2 == 0)
            {
                printf("%c", a[i][j]);
            }
        }
    }
}
int main(void)
{
    char arr[2][100] = {"pg-dac", "pg-dmc"};
    fun(arr, 2);
    return 0;
}
```

A. p-a
B. g-d
C. p p
D. error

Answer: A

4.

Find the output of the following code:

```
#include <stdio.h>
void fun(char a[][9], int rows)
{
    int i;
    for (i = 0; i < rows; i++)
    {
        printf("%c", *(*(a + i) + i));
    }
}
int main(void)
{
    char arr[2][9] = {"precat", "cdac"};
    fun(arr, 2);
    return 0;
}
```

A. pd
B. ea
C. pc
D. tc

Answer: A

5.

Find the output of the following code:

```
#include <stdio.h>
void fun(char a[][9], int r)
{
    for (int i = 0; i < r; i++)
    {
        printf("%c", a[i][strlen(a[i]) - 1]);
    }
}
int main(void)
{
    char arr[2][9] = {"PUNE", "KARAD"};
    fun(arr, 2);
    return 0;
}
```

A. ED
B. PK
C. NO OUTPUT
D. EK

Answer: A

1.

What happens if we try to access argv[argc]?

A. Array bounds to be handled by programmer
B. May Cause compile Time Error
C. Returns the last argument
D. Returns an empty string

Answer: A

2.

Find the output of the following code:

```
#include <stdio.h>
int main(int argc, char **argv)
{
	char **argptr = argv; int i=0;
	for (i ; *(argptr+i) != NULL ? printf("%s ",*(argptr+i)) : 0; i++);
	return !!argc - 1 + 1 - 1;
}
```

For above the program when we pass following command line argument what will be the output
".\a.exe Hello World"

A. Hello World
B. .\a.exe Hello World
C. Hello World ./a.exe
D. a.exe Hello World

Answer: B

3.

when we run this program without passing any command line argument then what will be the output for following the code ?

```
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *argv[])
{
	int sum = 0;
	for (int i = 1; i < argc; i++)
	{
		sum += atoi(argv[i]);
	}
	printf("%d", sum);
	return 0;
}
```

A. 5
B. 7
C. 0
D. Compilation error

Answer: C

4.

Which function is used to convert a command-line argument from string to integer?

A. itoa()
B. int()
C. atoi()
D. strtoint()

Answer: C

5. Which loop prints all the command-line arguments (excluding the program name)?

```
#include<stdio.h>
int main(int argc, char *argv[])
{
    // _______
        printf("%s\n", argv[i]);
    return 0;
}
```

A. for(int i = 0; i < argc; i++)
B. for(int i = 1; i < argc; i++)
C. for(int i = 1; i <= argc; i++)
D. for(int i = 0; i <= argc; i++)

Answer: B

6. Find the output of the following code:

```
#include <stdio.h>
int main(int argc, char *argv[])
{
    printf("%d\n", *(*(argv + 2) + 1) - *(*(argv + 1) + 1));
    return 0;
} //give Command line Argument as follows  "./a.exe abc bcd"
```

A. 1
B. 0
C. -1
D. 2

Answer: A

1. Find the output of the following code:

```
#include <stdio.h>
#define VAL 10
int main(void)
{
    #ifdef VAL
        printf("Yes");
    #else
        printf("No");
    return 0;
}
```

A. Yes
B. No
C. Nothing
D. Compilation error

Answer: D

2. Find the output of the following code:

```
#include <stdio.h>
#if 1
int main(void)
{
    printf("OK");
    return 0;
}
#else
    #error "Should not compile"
#endif
```

A. OK
B. Should not compile
C. Compilation error
D. Warning only

Answer: A

3.

Find the output of the following code:

```
#include <stdio.h>
#define VAL 5
#define THRESHOLD 3
#define STR_HIGH "High"
#define STR_LOW "Low"
#if VAL > THRESHOLD
	#if VAL % 2 == 1
		#define RESULT STR_HIGH
	#else
		#define RESULT "Medium"
	#endif
#else
		#define RESULT STR_LOW
#endif
int main(void)
{
	printf("%s", RESULT);
	return 0;
}
```

A. High
B. Low
C. RESULT
D. Compilation error

Answer: A

4.

Find the output of the following code:

```
#include <stdio.h>
#define MODE 1
#if MODE == 1
	#define VALUE 100
	#undef MODE
#else
	#define VALUE 200
#endif
#ifndef MODE
	#define MODE 2
#endif
int main(void)
{
	printf("%d %d", VALUE, MODE);
	return 0;
}
```

A. 200 1
B. 100 2
C. 100 1
D. Compile-time error

Answer: B

4.

Find the output of the following code:

```
#define DEBUG
#ifndef DEBUG
	#define PRINT(x) printf("DEBUG: %s\n", x)
#else
	#define PRINT(x)
#endif
```

```
int main(void)
{
    PRINT("Test");
    return 0;
}
```

A. DEBUG: Test

B. Test

C. No output

D. Compilation error

Answer: C

6. Find the output of the following code:

```
#include <stdio.h>
#define A 1
#def ine TOGGLE(x) x + 1
#ifdef A
    #undef A
    #define A TOGGLE(1)
#endif
#ifndef B
    #define B A
#endif
int main(void)
{
    printf("%d", B);
    return 0;
}
```

A. 1

B. 2

C. 3

D. Compile-time error

Answer: B

1. Find the output of the following code:

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>
void p(char *s)
{
    *s ? (printf("%c ", *s), p(s + 1)) : 0;
    return;
}
int main(void)
{
    char *n = (char *) malloc(6 * sizeof(char));
    strcpy(n, "HELLO");
    p(n);
    free(n);
    return 0;
}
```

A. H E L L O

B. HELLO

C. O L L E H

D. H E L L O followed by a segmentation fault

Answer: A

2. What is a memory leak?

A. Accessing invalid memory

B. Programmer do not free dynamically allocated memory

C. Deallocates a memory which has been already deallocated

D. Using uninitialized pointers

Answer: B

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3.

Find the output of the code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main(void)
{
    char *name = (char*)malloc(10*sizeof(char));
    strcpy(name, "Hello");
    printf("Before free: %s\n", name);

    char modifiedName[10];
    strcpy(modifiedName, name);
    free(name);

    modifiedName[0] = 'h';
    printf("After modification: %s\n", modifiedName);

    return 0;
}
```

A. Before free: Hello
After modification: Hello
B. Before free: Hello
After modification: hello
C. Before free: hello
After modification: hello
D. Undefined behavior

Answer: B

4.

If malloc() fails to allocate memory dynamically, it returns __.

A. -1
B. Crashes the program
C. NULL
D. 0
E. Both C and D

Answer: E

5.

```
int *ptr = malloc(5 * sizeof(int));
*(ptr + 5) = 42;
```

What is the result of the code above?

A. It correctly assigns 42 to the sixth element of the array
B. It causes a compiler error due to invalid pointer arithmetic
C. It compiles and runs, but the behavior is undefined
D. The code will always crash at runtime

Answer: C

1.

Find the output of the following code:

```
#include<stdio.h>
struct a
{
	int x;
};
struct b
{
	struct a i;
};
int main(void)
{
	struct a o = {5};
	struct b p = {55};
	p.i=o;
	printf("%d\n", p.i.x);
	return 0;
}
```

A. 55

B. 5

C. Compiler error

D. 0

Answer: B

2.

Find the output of the following code:

```
#include <stdio.h>
struct Point
{
	int x, y;
};
int main(void)
{
	struct Point arr[] = { {10, 20}, {30, 40}, {50, 60} };
	struct Point *p = arr + 1;
	printf("%d %d %d\n", (p-1)->x, (p - 2)->y, (++p)->y);
	return 0;
}
```

A. 30 20 60

B. 30 60 40

C. 10 60 20

D. 30 40 60

Answer: A

3.

Find the output of the following code:

```
#include <stdio.h>
struct S {
	int a;
};
int main(void)
{
	struct S s1 = {5};
	struct S *p1 = &s1;	struct S *p2 = p1;
	p2->a = 10;
	printf("%d\n", s1.a);
	return 0;
}
```

A. 5

B. 10

C. Compiler error

D. Garbage

Answer: B

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4. Find the output of the following code:

```
#include <stdio.h>
struct S
{
	unsigned int x : 4;
};
int main(void)
{
	struct S s = {16};
	printf("%u\n", s.x);
	return 0;
}
```

A. 16
B. 0
C. 4
D. Undefined

Answer: B

5. Find the output of the following code:

```
#include <stdio.h>
struct S
{
	signed int a: 3;
};
int main(void)
{
	struct S s;
	s.a = 13;
	printf("%d\n", s.a);
	return 0;
}
```


A. 13
B. -3
C. 5
D. Undefined behavior

Answer: B

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1.

Find the output of the following:

```
#include <stdio.h>
int main(void)
{
    FILE *fp = fopen("abc.txt", "w+");
    if (!fp)
    {
        perror("File open error");
        return 1;
    }
    fprintf(fp, "123");
    fseek(fp, 0, SEEK_SET);
    fprintf(fp, "456");
    fflush(fp);
    fseek(fp, 0, SEEK_SET);
    char buffer[10];
    fgets(buffer, sizeof(buffer), fp);
    buffer[4] = '\0';

    printf("File content: %s\n", buffer);
    fclose(fp);
    return 0;
}
```

A. File content: 123456

B. File content: 456

C. File content: 123

D. File content: Undefined

Answer: B

2.

Find the output of the following:

[Consider the contents of the "abc.txt" as "sunbeam pune karad"]

```
#include <stdio.h>
int main(void)
{
    FILE *fp = fopen("abc.txt", "r");
    char *p = NULL;
    char c;
    int count = 0;
    while ((c = fgetc(fp)) != EOF)
    {
        if (count % 2 == 0)
        {
            p = &c;
            printf("%c", *p);
        }
        count++;
    }
    fclose(fp);
    return 0;
}
```

A. snempn aa

B. Until EOF

C. Just 1 char

D. Infinite loop

Answer: A

3. Find the output of the following:

```
#include<stdio.h>
int main(void)
{
    FILE *fp = fopen("abc.txt", "w");
    fputc('X', fp);
    fclose(fp);
    fp = fopen("abc.txt", "r");
    char ch = fgetc(fp);
    printf("%c", ch);     fclose(fp);
    return 0;
}
```

A. Compile error
B. X
C. Garbage
D. EOF

Answer: B

4. Find the output of the following:

```
#include<stdio.h>
int main(void)
{
    FILE *fp = fopen("abc.txt", "w");
    fprintf(fp, "%c", 'Z');
    fclose(fp);
    fp = fopen("abc.txt", "r");
    printf("%d", fgetc(fp));
    return 0;
}
```


A. 90
B. Z
C. 'Z'
D. Compiler error

Answer: A

5. Find the output of the following:
[Consider the contents of the “file.txt” as “1234567890”]

```
#include<stdio.h>
int main(void)
{
    FILE *fp = fopen("file.txt", "w+");
    fprintf(fp, "abc");
    fseek(fp, 0, SEEK_SET);
    fprintf(fp, “%s”, "1234567890");
    fseek(fp, 0, SEEK_SET);
    char buffer[10];
    fgets(buffer, sizeof(buffer), fp);
    buffer[5] = '\0';
    printf("%s\n", buffer);
    fclose(fp);
    return 0;
}
```

A. 123
B. 12345
C. 45673
D. 345

Answer: B

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